
Understanding topic-based social media data on water

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Abstract

Social media platforms provide a large-scale and continuously updated source of public discourse that can be used to study societal perceptions of environmental issues. This project develops a data processing pipeline to collect and analyze water-related content from BlueSky and Mastodon using a hierarchical set of keywords. The collected posts and metadata are stored in a structured database to support downstream analysis.

The workflow includes data collection, preprocessing, sentiment analysis, and temporal trend analysis to examine how attention and sentiment toward water-related topics evolve over time. While the analytical component is still in progress, the framework provides a scalable basis for monitoring environmental discourse online and supporting future research in environmental analytics and social sensing.

1 Introduction

Social media platforms provide a rich and continuously updated source of public discourse, making them valuable for understanding societal attitudes toward environmental and resource-related topics. Water-related issues, including consumption, scarcity, pollution, and policy debates, are increasingly reflected in online discussions.

This project aims to build and analyze a structured pipeline for extracting insights from water-related social media content. The workflow consists of data scraping, database structuring, preprocessing and cleaning, sentiment analysis, and temporal trend analysis. By combining large-scale data collection with natural language processing techniques, the study seeks to identify how sentiment and attention toward water-related topics evolve over time and across different linguistic and thematic contexts.

The results provide a foundation for monitoring public perception and can support future work in environmental analytics, policy evaluation, and social sensing systems.

2 Data extraction

The data was collected using the free APIs of the text-based social media platforms BlueSky and Mastodon. Given a hierarchy of water-based keywords from (CITE SOURCE), we run a collection pipeline that searches the platforms using these keywords, scrapes the posts and their metadata from the search results, and saves them to a database schema adapted from (SOURCE JULIA). This pipeline runs weekly on a subset of frequent keywords from the hierarchy.¹



Figure 1: Average keyword sentiment over time for 78,000 sentiment-scored posts. We only plot the top 15 most frequently used keywords.

3 Data analysis

Given the database of scraped posts from the two platforms, we run a BERT-based sentiment analysis model on the post contents for each keyword. This information allows us not only to examine keyword-specific sentiments, but also overall sentiment trends over time.

Figure 2 places the posts into sentiment buckets of size 0.2, allowing us to examine trends for various sentiment intervals. The top graph shows the mean number of likes per sentiment bucket. Over the x-axis we show the amount of posts in each bucket for clarity. Clearly, the majority of posts fall into the middle buckets, indicating neutral sentiments. As expected, polarizing posts (i.e., posts that are predicted as being either very positive or very negative) tend to have higher mean interactions than neutral ones. Interestingly, the most negative bucket has the highest mean number of likes out of all intervals.

The heatmap breaks down the keyword share for each of these sentiment buckets. To remove clutter, we only plot the top 15 most frequent keywords. For each keyword, we see the number of posts falling into each bucket as a fraction of the overall posts for that keyword (the percentage) as well as the absolute number of posts in parentheses. A more intensely coloured cell indicates that a higher fraction of that keyword’s posts fall into that sentiment bucket.

4 Limitations

Gallegos et al. [2024] list several shortcomings of the most widespread bias evaluation benchmarks. On the one hand, many metrics assume naive notions of bias, have only weak or inconsistent relationships with biases in downstream tasks [Kaneko et al., 2022, Cao et al., 2022, Cabello et al., 2023], or are drastically affected by parameter choices to the point of contradicting results

¹This is due to the fact that many of the keywords provided very little results, making meaningful analysis difficult in later steps.

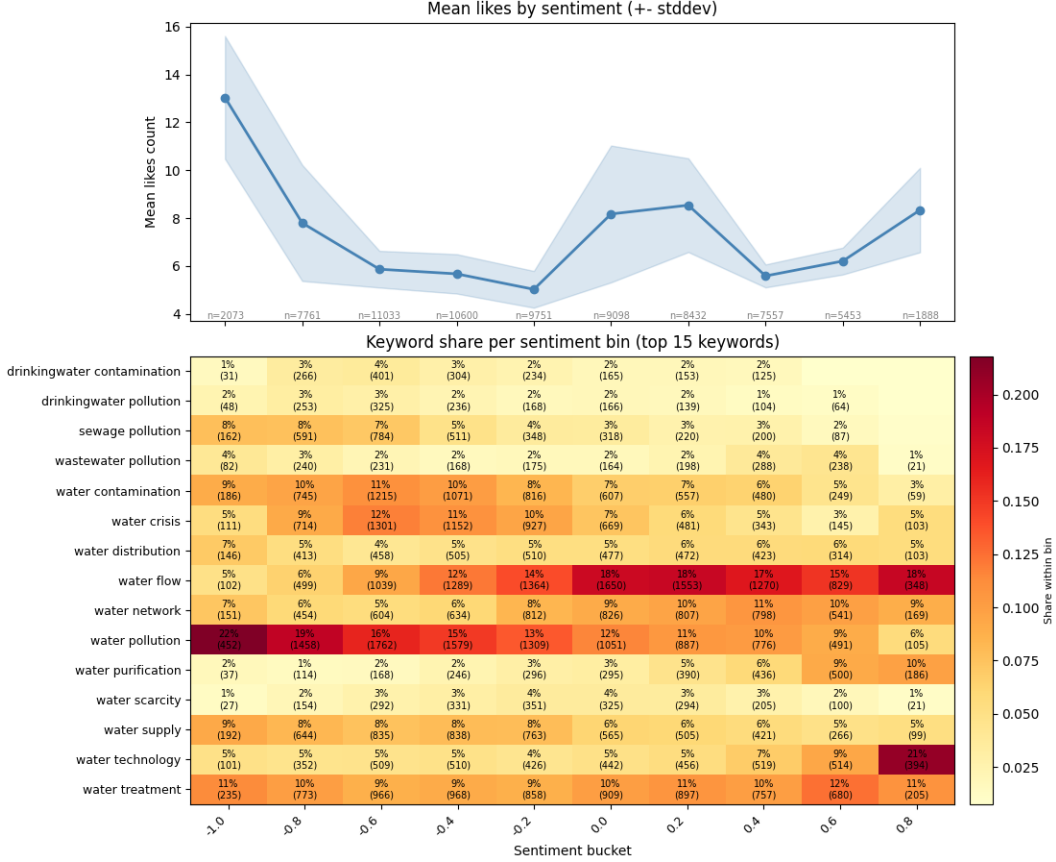


Figure 2: Keyword share for 78,000 sentiment-scored posts. -1 corresponds to the most negative sentiment score, +1 to the most positive sentiment score. We only plot the top 15 most frequently used keywords.

[Akyürek et al., 2022]. On the other hand, many of the well-known datasets may be limited in their generalizability, given that many are constructed from the context of the United States, yet are offered as benchmarks for English language models everywhere. Prominent coreference resolution-based evaluation benchmarks and their datasets have been called into question by Selvam et al. [2023], who observe that shallow sentence modifications that retain social biases can drastically affect bias scores.

Similarly, mitigation techniques at all stages of the LLM pipeline each have their drawbacks. In pre-processing for example, replacing or masking examples can obscure relevant power imbalances, while unbiased or diversified prompts have been shown to lead to similar outputs [Borchers et al., 2022, Li et al., 2024].

Resnik [2025] more fundamentally criticizes LLM bias mitigation in general, primarily focusing on reinforcement learning with human feedback as what is termed the dominant approach for guiding LLMs to align with human goals such as fairness [Casper et al., 2023]. The most fundamental issue being pointed out is the notion that this approach essentially replaces one set of black-box biases, stemming from a huge amount of human-generated language data, with another set of biases based on a small number of human reviewers. Furthermore, [Resnik, 2025] makes the claim that this process is navigating a tradeoff between pushing the model away from latent representational structure that contains harmful biases, whilst keeping it faithful to the enormous amount of data on which it was resource-intensively trained – a process taking place in a sea of uncertainty without a clear understanding of how the underlying structure is being affected.

5 Conclusion

Questioning the methodologies and navigating the inherent challenges of bias research with LLMs is an essential task – Blodgett et al. [2020] notably propose grounding bias research in the relevant research outside of NLP, as well as being forthright with respect to the normative reasoning on which the conducted research is founded. Moreover, the challenge of navigating fairness in LLMs demands steps that go beyond developing systems to recognize and mitigate biases, that is, critically engaging with the systems of power that lead to these undesirable outcomes in the first place.

The development of systems centered around operationalizing topics as contextual and shifting as bias and fairness is necessarily political; highlighting this issue, Bender et al. [2021] remark that algorithmic definitions of what constitutes inappropriate communication, for example, will align with some views and not with others.² From this follows that any attempt to measure biases encoded into systems is by its very nature political and must be done in relation to particular contexts and marginalized perspectives.

When it comes to bias evaluation and mitigation, Resnik [2025] posits that there are no theory-free methods, and philosophical commitments are unavoidable in any attempt to model human behavior. Extending this perspective, Green [2019] points out that the underlying values and assumptions of computer science itself tend to be those of the dominant social groups, a sentiment also reflected by Kalluri [2020]. There is no neutral standpoint from which science proceeds. As Daniel C. Dennett observed, “there is no such thing as philosophy-free science; there is only science whose philosophical baggage is taken on board without examination” [Dennett, 1996].

²This can be seen when observing online harassment, for example; research shows that even those experiencing these harms have varying definitions of the term Duggan [2017]

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